

Lucas WÄLTI

PhD Student, DISAL, EPFL
Robotics Engineer, EPFL M.Sc. in Robotics

[in linkedin.com/in/lucas-waelti-30046b185](https://www.linkedin.com/in/lucas-waelti-30046b185) github.com/LucasWaelti gitlab.com/LucasWaelti
☎ +41 78 318 75 89 @ lucas.waelti@gmail.com 📍 Lausanne, Switzerland



PERSONAL INFORMATION

PhD student at the Distributed Intelligent Systems and Algorithms Laboratory (DISAL), supervised by Prof. Alcherio Martinoli. My research focuses on the inspection of infrastructural assets with multiple autonomous UAVs, applied to steel infrastructures. My research interests include autonomous robotics, navigation, perception, multi-robot systems, embedded systems, simulation, and machine learning. [DISAL contact page](#).

SKILLS

Knowledge in	Robotics, Unmanned Aerial Vehicles (UAVs), Multi-Robot Systems, Simulation, Robot Operating System (ROS), System Control, Sensor Fusion, Embedded Systems, Deep Learning, Deep Reinforcement Learning, Computer Vision, 3D CAD
Programming	C/C++, Python, bash, Matlab, Java
Simulation	Robot modeling and control: Webots, Gazebo
Software/Tools	Linux (Ubuntu), Windows, git, Webots, ROS, PX4, PyTorch, Pandas, Matlab, Docker, FreeCAD, 3D printing

PROFESSIONAL EXPERIENCE

Now December 2021	PhD student, DISTRIBUTED INTELLIGENT SYSTEMS AND ALGORITHMS LABORATORY, EPFL, Switzerland Research topic: <i>Distributed Robotic Inspection of Infrastructural Assets</i> . Thesis director: Prof. Alcherio Martinoli The application of autonomous UAVs for the inspection of steel infrastructures is investigated, with a focus on the navigation challenges. This use-case requires the combination of adequate state estimation, object detection, path planning and disturbance detection methods, on top of a multi-robot coordination scheme. Practical experiments are typically carried out with the Starling MAV , produced by ModalAI, Inc.
November 2021 October 2020	R&D Engineer, DISTRIBUTED INTELLIGENT SYSTEMS AND ALGORITHMS LABORATORY, EPFL, Switzerland Full time engineer working in the continuity of my master thesis. Hired as a scientific assistant to further develop the autonomous inspection functionalities of the drone for elevator shafts. The improvements include the embedding of supplementary sensors (such as range finders over I ² C) and various updates in navigation functionalities. ROS C/C++ Python Webots docker embedded systems yocto PX4
August 2020 February 2020	Master Thesis, SCHINDLER NEW TECHNOLOGIES OFFICE, Switzerland Master Thesis in the industry. Implementation of an autonomous drone for elevator shafts inspection. Illustrative video: https://youtu.be/6SZQSiVCu-U ROS C/C++ Python Webots docker embedded systems yocto PX4
May 2018 January 2018	Military Service, SWISS AIR FORCES, Switzerland Formation in the context of the mandatory military service as a <i>soldier</i> and <i>mechanic</i> on fighter aircrafts F/A-18 Hornet C/D.
December 2017 October 2017	Engineering Internship, CYBERBOTICS, Switzerland Adaptation of a virtual bipedal robot trained using deep reinforcement learning (cf. DeepLoco). The virtual robot first had to be modeled in Webots. The original controller had to be then adapted to work in the Webots simulation environment. C/C++ Webots

LANGUAGES

French	Mother tongue
English	Fluent, C1-C2
German	Good level, C1

DIPLOMAS

- Master of Science M.Sc. in Robotics, EPFL, 2020
- Bachelor of Science B.Sc. in Microengineering, EPFL, 2017
- Baccalaureate, Gymnasium Auguste Piccard, 2014
- Goethe-Zertifikat C1 (German), Goethe-Institut e.V., 2014

PUBLICATIONS

- L. Wälti, I. Kagan Erunsal and A. Martinoli, "**Multi-Robot Online Coverage with a Team of Resource-Constrained Micro Aerial Vehicles**", 2024 Distributed Autonomous Robotic Systems (DARS), New York, USA, 2024.
- L. Wälti and A. Martinoli, "**Lumped Drag Model Identification and Real-Time External Force Detection for Rotary-Wing Micro Aerial Vehicles**," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 3853-3860, DOI: [10.1109/ICRA57147.2024.10610862](https://doi.org/10.1109/ICRA57147.2024.10610862).

PATENTS

- [WO2023104494A1](#), "**Method for controlling a drone that can fly in a lift shaft of a lift system and lift system inspection assembly**", inventors: Raphael BIZTI, Alberto CHIAPPA, Martin KUSSEROW, Lucas WÄLTI

ATTENDED CONFERENCES AND EVENTS

- [2024 Distributed Autonomous Robotic Systems \(DARS\)](#) in New York, USA, November 28th to 30th, 2024
- [2024 IEEE International Conference on Robotics and Automation \(ICRA\)](#) in Yokohama, Japan, May 13th to 17th, 2024
- [2023 IEEE Summer School on Multi-Robot Systems and UAVs](#) in Prague, Czech Republic, July 3rd to 7th, 2023